



Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University

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College of Informatics and Computing Sciences

CC 100 – INTRODUCTION TO COMPUTING

Information Technology and Computer Science Department

Academic Year 2026-2027

I. Key Terminology & Vocabulary

- **Algorithm**: A step-by-step set of instructions or computational rules designed to perform a specific task or solve a problem.
- **Architecture**: The structural blueprint and functional design of a computer system, detailing how hardware components interact (e.g., Von Neumann Architecture).
- **Arithmetic/Logic Unit (ALU)**: The core component of the CPU that performs arithmetic calculations (addition, subtraction) and logical evaluations (AND, OR, NOT).
- **Bit & Byte**: A **bit** (binary digit) is the fundamental unit of digital data, taking a value of either 0 or 1. A **byte** consists of 8 bits.
- **Central Processing Unit (CPU)**: The "brain" of the computer that interprets and executes system and application instructions.
- **Cloud Computing**: Delivery of computing services (storage, databases, software) over the internet, allowing remote data access, syncing, and collaboration.
- **Control Unit (CU)**: The internal CPU module that extracts instructions from memory, decodes them, and manages execution flow across hardware.
- **Data Center**: A dedicated facility housing connected server hardware used to process, store, and distribute massive datasets for cloud services and platforms.
- **Disk Operating System (DOS)**: An early text-based, command-line interface operating system developed by Microsoft for IBM PCs.
- **Graphical User Interface (GUI)**: A visual interface featuring icons, windows, menus, and pointers, enabling user navigation via pointing devices rather than text commands.
- **Information Technology (IT)**: The application of computer systems, networking, storage, and software to process, retrieve, and safeguard digital information.
- **Mainframe Computer**: A massive, centralized computing system used historically (and by large enterprises) for bulk data processing and high-volume transactions.
- **Microcomputer**: A compact, single-user computer built around a microprocessor chip (e.g., desktops, laptops, smartphones).
- **Motherboard**: The main printed circuit board (PCB) that holds the CPU, memory, expansion slots, and bridges physical communication between peripherals.
- **Operating System (OS)**: The master system software that manages hardware resources, allocates memory, handles file systems, and provides an execution platform for applications.
- **Random Access Memory (RAM)**: Fast, volatile main memory used by the CPU to read and write temporary operational data. Data is lost when power shuts down.
- **Read-Only Memory (ROM)**: Non-volatile storage containing permanent, fundamental startup instructions (e.g., BIOS/UEFI) required to boot up the system.
- **Solid-State Drive (SSD) vs. Hard Disk Drive (HDD)**:
 - **HDD**: Secondary magnetic storage with moving mechanical platters; slower and cheaper per gigabyte.



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- **SSD**: Secondary flash storage using transistors with no moving mechanical parts; durable, power-efficient, and drastically faster.
- **System Bus**: A shared electrical communication pathway on the motherboard that transports data and control signals between the CPU, RAM, and internal devices.
- **Transistor**: A microscopic electronic semiconductor component functioning as a fast binary switch (0 or 1) within microprocessors.
- **Unix**: A multi-user, multitasking operating system built in the late 1960s that serves as the architectural predecessor for Linux, macOS, Android, and iOS.
- **Von Neumann Architecture**: A foundational design model for electronic computers featuring a processing unit (ALU and Control Unit), memory unit, and Input/Output mechanisms.

II. Historical Milestones & Key Timeline

- **1820s** — **Analytical Engine Design: Charles Babbage** designs the earliest mechanical computer framework, establishing concepts for memory storage and processing. **Ada Lovelace** writes algorithms for it, earning her recognition as the world's first computer programmer.
- **1945** — **Von Neumann Architecture: John von Neumann** formulates the electronic computer architecture where program instructions and data reside in a shared memory space.
- **1946** — **ENIAC Unveiled**: The **Electronic Numerical Integrator and Computer** is completed for the US Army. It becomes the world's first general-purpose, fully electronic, digital programmable computer.
- **1956** — **AT&T Monopoly Agreement**: A legal consent decree prevents AT&T from commercializing non-telecom inventions, forcing them to license Unix source code freely to academic researchers.
- **1968** — **"Mother of All Demos"**: **Douglas Engelbart** demonstrates key personal computing capabilities, including the mouse, networking, hypertext, video conferencing, and word processing.
- **1969** — **Unix Development**: **Ken Thompson** and **Dennis Ritchie** create Unix at AT&T Bell Labs, originally written to host games like *Space Travel*.
- **1981** — **Standard Networking Protocols & IBM PC**: Networking standards consolidate, while the IBM PC emerges with Microsoft's DOS, launching the personal computer wave.
- **1985** — **Release of Microsoft Windows**: Microsoft introduces Windows (a GUI operating environment for DOS) alongside spreadsheet applications like Excel.
- **1990s** — **Modern Internet & PC Adoption**: Desktop computers become widespread household appliances. The World Wide Web bridges globally connected networks.
- **2007** — **Modern Mobile Revolution**: Apple introduces the iPhone, uniting multi-touch interfaces, mobile internet, and cellular application stores into consumer devices.

III. Notable Historical & Key Tech Figures

- **Charles Babbage**: "Father of Computing" who conceptualized mechanical calculating engines (Difference Engine & Analytical Engine).
- **Ada Lovelace**: Mathematician who documented Babbage's work and formulated the first machine-oriented algorithm.



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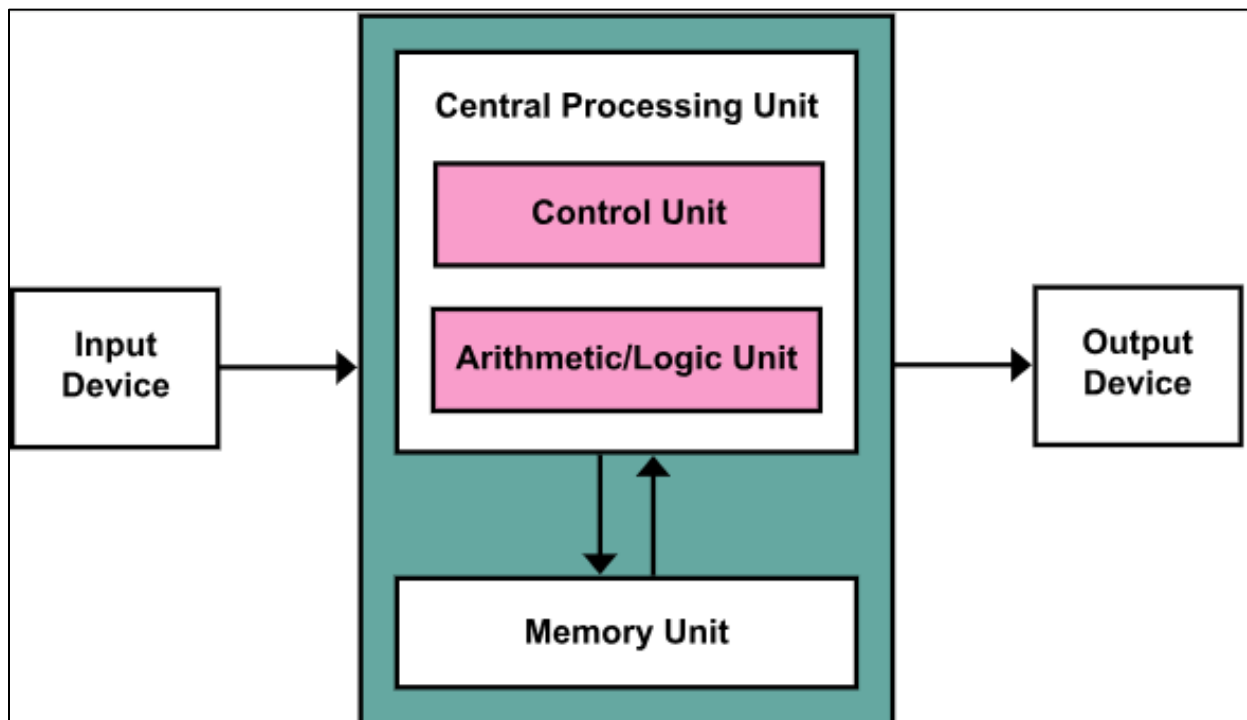
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- **ENIAC Programmers (The ENIAC Six)**: A team of women programmers (including Kay McNulty, Betty Jennings, Betty Snyder, Marlyn Wescoff, Frances Bilas, and Ruth Lichterman) who manually programmed ENIAC via plugboards and function tables.
- **John von Neumann**: Mathematician and physicist who documented the fundamental CPU-Memory-I/O structural paradigm.
- **Douglas Engelbart**: Computing pioneer who led the 1968 "Mother of All Demos" introducing mice, word processing, and early interactive displays.
- **Ken Thompson & Dennis Ritchie**: Computer scientists at AT&T Bell Labs who developed the Unix operating system. Dennis Ritchie also created the C programming language.
- **Doug Klunder & Charles Simonyi**: Software engineers at Microsoft who led the core development of Microsoft Excel (Klunder) and Microsoft Word (Simonyi).
- **Steve Jobs**: Co-founder of Apple Inc. who unveiled key consumer products including the Macintosh (1984) and the iPhone (2007).

IV. Hardware Systems & Von Neumann Architecture

The physical equipment of a computing device operates on a continuous fetch-decode-execute cycle via the Von Neumann model:



- **CPU**: Handles computation. Divided into the **Control Unit** (directs operation sequences) and **Arithmetic/Logic Unit** (math/logical steps).
- **GPU (Graphics Processing Unit)**: Specialized coprocessor with thousands of smaller cores executing parallel workloads (graphics rendering, ML processing, video encoding).
- **Motherboard**: The main circuit interconnect board running system buses between CPU sockets, RAM slots, storage interfaces, and expansion ports.



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V. Input & Output (I/O) Hardware

In computer systems, I/O devices act as the essential interface bridging users and secondary environments with the CPU and system memory.

Input Devices

Converts human actions or real-world physical data into binary signals (0s and 1s) for system processing:

- **Keyboard**: Primary text/command interface using switches to map keys into character codes.
- **Pointing Devices**: Mouse, trackpad, and stylus devices capturing motion across spatial coordinates.
- **Optical & Imaging**: Image scanners, barcode/QR readers, webcams, and digital cameras.
- **Audio Inputs**: Microphones utilizing Audio Digital Converters (ADC) to transform sound into digital audio.
- **Biometric Sensors**: Fingerprint scanners, facial cameras, and iris sensors for authentication.

Output Devices

Translates processed binary data into human-perceptible media or physical actions:

- **Display Monitors**: LCD, LED, and OLED screens rendering visual interfaces mapped by the GPU.
- **Printers & Plotters**: Inkjet (liquid ink), Laser (heat-fused toner), and Vector Plotters (architectural blueprints).
- **Audio Outputs**: Speakers and headphones using Digital-to-Analog Converters (DAC) to play audio.
- **Haptic Feedback**: Mechanical vibration actuators providing tactile responses.

VI. Storage Metrics & Data Measurement

Computing measurements operate on binary factors ($2^{10} = 1024$):

Memory unit	Description
Kilo Byte	1 KB = 1024 Bytes
Mega Byte	1 MB = 1024 KB
Giga Byte	1 GB = 1024 MB
Tera Byte	1 TB = 1024 GB
Peta Byte	1 PB = 1024 TB
Hexa Byte	1 EB = 1024 PB
Zetta Byte	1 ZB = 1024 EB
Yotta Byte	1 YB =1024 ZB
Bronto Byte	1 Bronto Byte = 1024 YB
Geop Byte	1 Geo Byte = 1024 Bronto Bytes



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VII. Operating Systems & File Management

Operation	Action & Description	Windows Shortcut	macOS Shortcut	Safety & Important Notes
Renaming	Changes the file’s label or file extension without modifying the internal file contents.	F2	Return (Enter)	Avoid changing the file extension (e.g., .docx to .pdf) manually, as it may render the file unopenable by its default program.
Copying	Creates an exact duplicate of a file or folder in a new location while preserving the original.	Ctrl + C → Ctrl + V	Cmd + C → Cmd + V	Requires enough free disk space in the destination drive to hold the duplicated data size.
Moving	Transfers a file or folder to a new folder path, deleting it from the source location once copied.	Ctrl + X → Ctrl + V	Cmd + C → Cmd + Option + V	If moved across different physical drives, the system performs a copy-then-delete action automatically.
Deleting	Removes a file from its current directory path and sends it to a temporary staging area (Recycle Bin / Trash).	Delete	Cmd + Delete	Permanent deletion (Shift + Delete on Windows) bypasses the staging area and immediately marks disk sectors as free space.
Restoring	Recovers a deleted file from the temporary staging area back to its original or designated directory path.	Open Recycle Bin → Right-Click → <i>Restore</i>	Open Trash → Right-Click → <i>Put Back</i>	Files cannot be restored via software if the Recycle Bin/Trash has been emptied or permanently purged.

VIII. IT Career Paths & Workplace Skills

- Specializations:** Software Engineering, Network Architecture, Cybersecurity Analysis, Computer Forensics, Cloud Engineering, and Data Science.
- Key Workplace Tiers:** Soft Skills (communication/collaboration), System Administration (OS setup/file organization), Productivity Suites (Office/Google Workspace), and Remote Collaboration Tools (Slack, Teams, Zoom).